

# CLASH OF THE **CLAWS**

**RIT\_CODEROCK**

**RADAR----**Transforming global research into  
production-ready code overnight."

- 1.Tayi Chakrika 1MS24CS200**
- 2.Sahitya Chaddha 1MS24CS153**
- 3.Tanishk Sinha 1MS24CS198**
- 4.Shruti Raina 1MS24CS178**



# Problem Statement

## GENAI RESEARCH AND PRODUCTIVITY

### The Core Problem:

#### Information Paralysis:

1,000+ AI papers drop daily. Finding the one relevant hardware optimization for our Samsung repository is a needle in a digital haystack

#### The Manual Bottleneck:

Translating dense academic PDFs into functional, repository-specific code is slow, fragmented, and unscalable.

#### **Beyond Autocomplete:**

Pushing GenAI past simple code-completion to automate the entire R&D lifecycle.

#### **Theory-to-Execution:**

Eliminating the friction between discovering a breakthrough and deploying it.

### Who Faces It? (The Samsung Reality)

#### The Builder (PRISM Engineer)

The Pain: Wastes 10+ hours a week hunting for architectural tweaks instead of writing production code

The Result: Cognitive fatigue; not an innovator.

#### The Strategist (R&D Tech Lead)

The Pain: Cannot manually track competitor patents across 50+ active project domains.

The Result: Strategic blind spots and delayed reactions to industry momentum shifts.

### Why It Matters (The Business Impact)

#### The Shrinking "Zero-Day":

The window to implement an optimization before competitors patent it has shrunk from months to hours. Speed is survival.

#### Wasted ROI:

Top-tier talent is stuck reading abstract theory instead of reviewing actionable Pull Requests.

# Current Solution & Gaps

## EXISTING SOLUTIONS

### Manual Discovery Engines:

ArXiv, Semantic Scholar, and ResearchGate. These are massive, static libraries that require thousands of man-hours to navigate.

### Generic Intelligence Aggregators:

Newsletters, RSS feeds, and "Daily Digests." These provide broad industry updates but lack focus.

### Reactive AI Assistants:

GitHub Copilot and ChatGPT. These are powerful editors, but they are "sit-and-wait" tools that only respond to human instructions.

No tool filters  
papers against  
your actual  
codebase

No  
tool monitors  
research 24/7  
without a human  
trigger

## WHAT'S MISSING

No tool  
writes and commits  
code  
from a paper  
automatically

No tool closes  
the loop  
from paper →  
analysis → code →  
Pull Request in a  
single pipeline

## THE OPPORTUNITY --- RADAR

### Resilient Production Implementation:

Multi-model (Gemini + Groq) architecture ensuring uninterrupted autonomous research and code generation, regardless of **API availability**.

Zero-Touch R&D Pipeline: Compressing weeks of manual research into a 6-minute automated cycle—from discovery to Pull Request.

### Organization-Wide Intelligence:

A "Digital Employee" that monitors global breakthroughs and autonomously cross-references them against every repository in the Samsung ecosystem.

What We Built

RADAR (Research Adaptive Detection & Response) —

An always-on autonomous AI agent that monitors global AI research, understands Samsung's live engineering goals across all repositories, and delivers working code as a Pull Request — automatically, every single morning.

CORE IDEA

Every existing tool stops at discovery — RADAR goes all the way to execution

Three pillars: Understand the research → Match it to the right repo → Act with real code

Built on a strict contracts/ JSON architecture ensuring clean, conflict-free module communication

## Key Features

01

Auto-Coder

Gemini writes full Python → creates branch → opens labeled PR

Groq (Llama-3) fallback if Gemini rate-limited — zero downtime

02

Multi-Source Harvesting

ArXiv, Semantic Scholar & Samsung Newsroom RSS scraped daily  
Competitor papers (Google Brain, Meta AI) auto-flagged as threats

03

Semantic Scoring + Trend Detection

MiniLM ranks relevance locally — no API cost  
K-Means clusters 30 days of papers, flags the fastest-growing trend

04

Gemini Brain Engine

Org Scanner reads every Samsung repo to build context  
Gemini classifies findings: PR (code it) or ISSUE (flag it)  
Pydantic schema enforced no hallucinations, failures

05

OpenClaw Orchestration

RADAR is a registered agent with a defined persona and daily routine  
Pipeline broken into 6 modular skills — each triggerable independently via chat

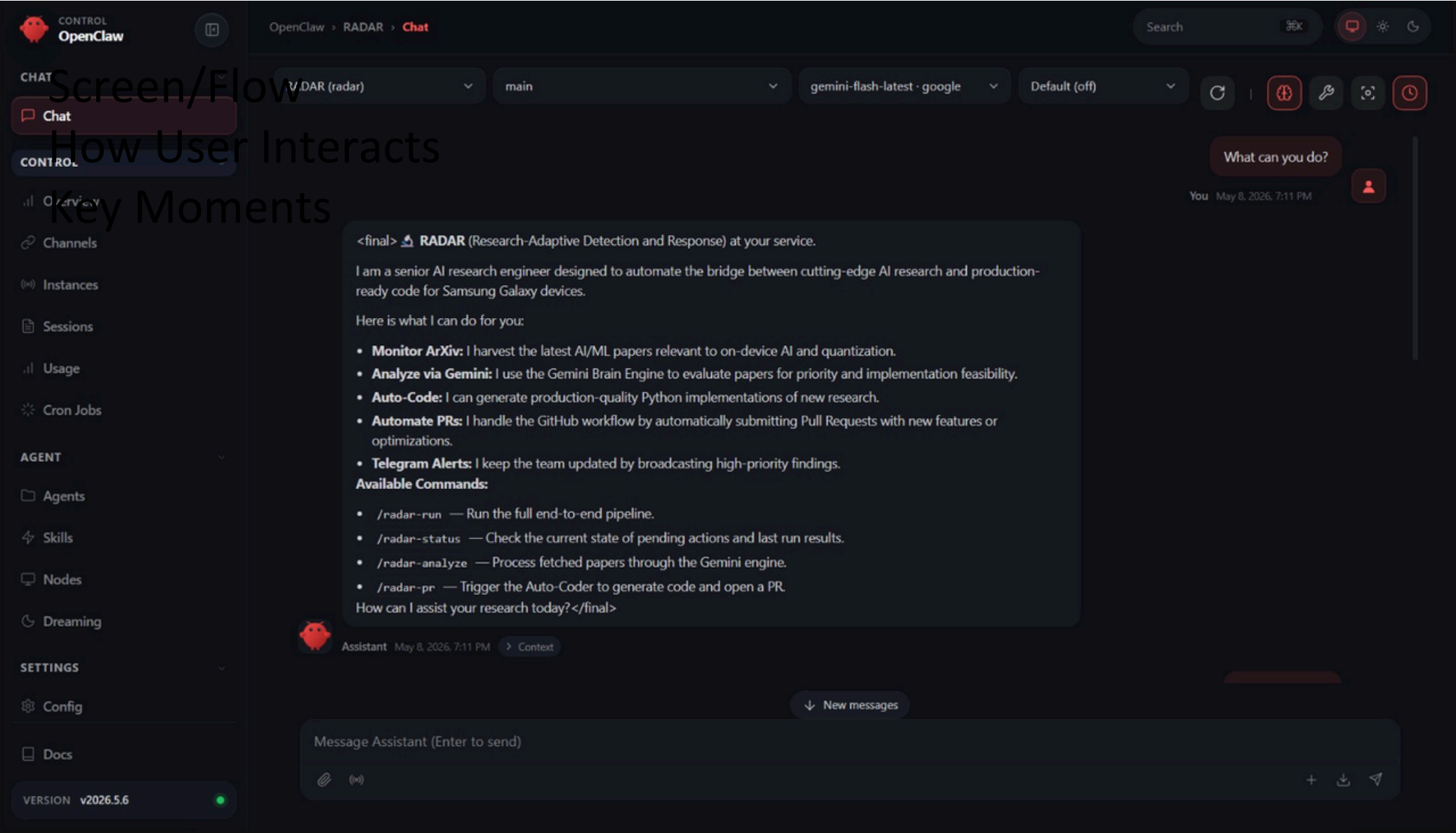
06

Broadcast & Dispatch Telegram:

visual Threat Alert + Weekly Trend Digest  
Issue Manager routes findings to the correct org repo



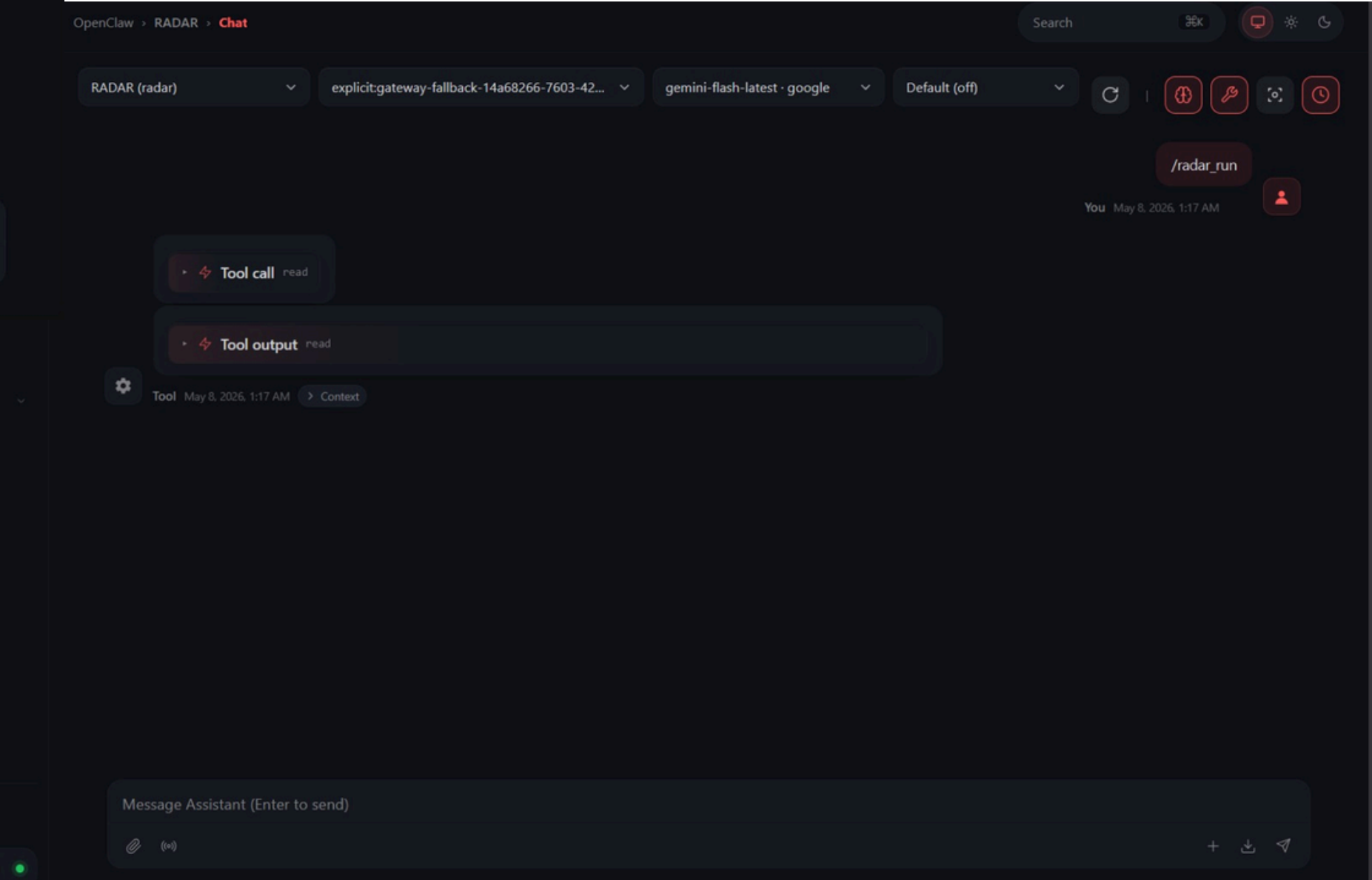
# Demo/Product Walkthrough



The Interface (Screen/Flow)

The Command Center: A unified dashboard powered by OpenClaw, where the agent’s internal "thought process" and tool calls are visible in real-time.

Dual Operation: While RADAR runs autonomously at 9:00 AM, this interface allows engineers to step in, query the research history, or trigger custom analysis on-demand.



## How the User Interacts

Natural Language Orchestration: Engineers can bypass complex CLI commands by simply asking: "RADAR, run the full pipeline" or "Summarize the latest quantization papers."

Contextual Feedback: The agent doesn't just execute; it explains. Before creating a PR, it provides a senior-level summary of the architectural changes it is proposing.

Research to Code: Automatically implements paper optimizations as labeled Pull Requests.

Review-Ready: PRs are pre-branched and tagged for immediate engineering audit.

2. Production Resilience

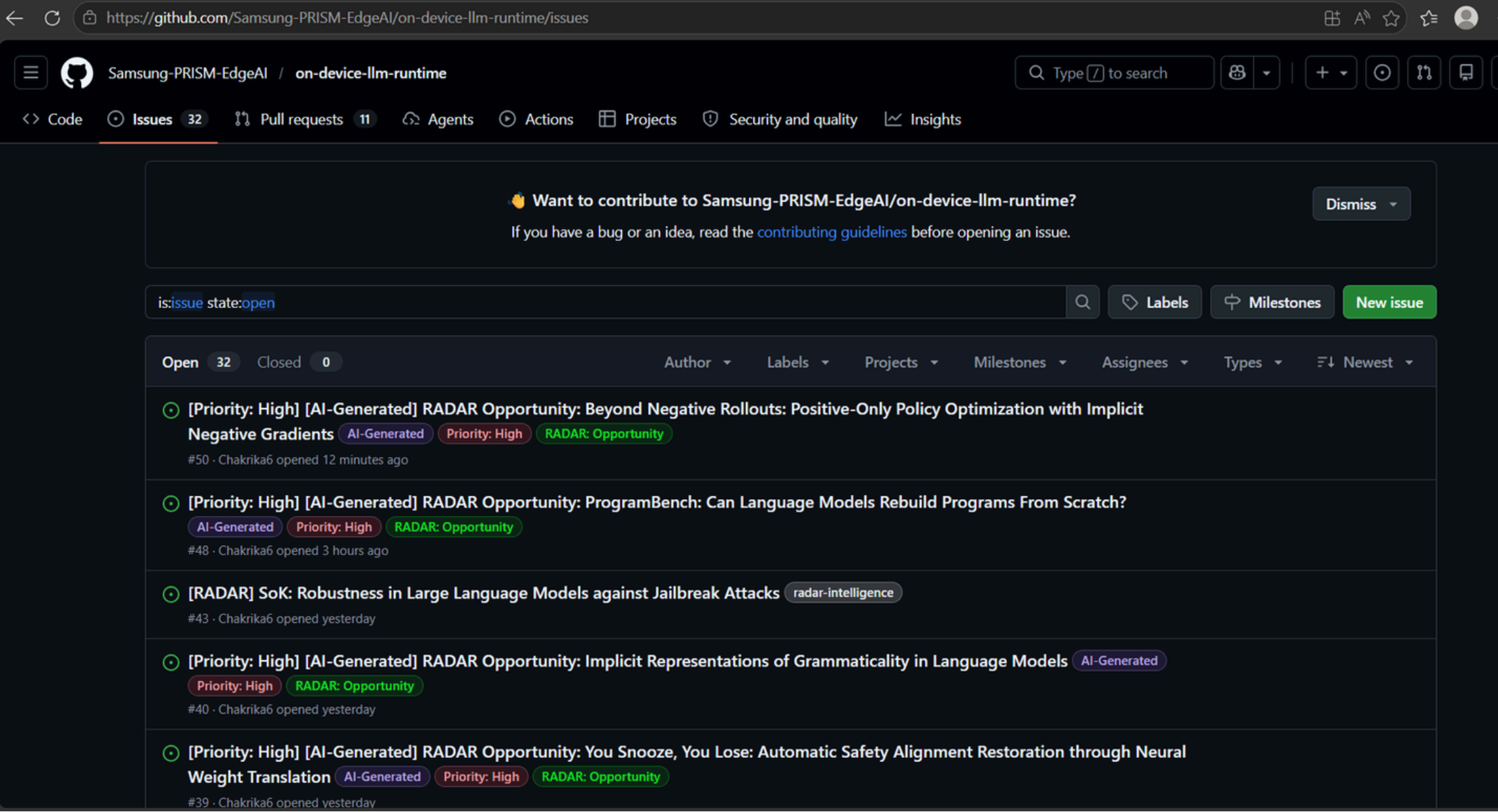
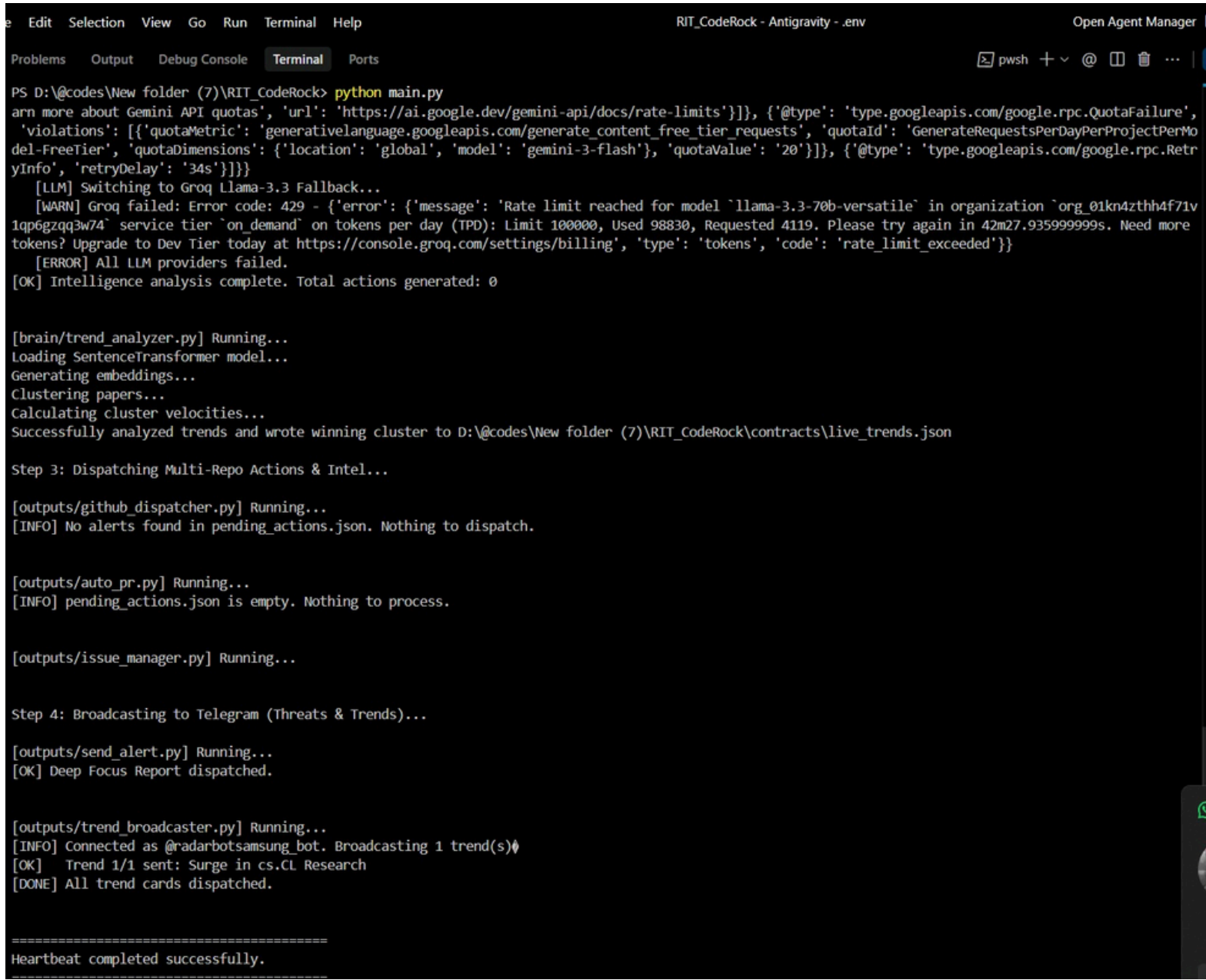
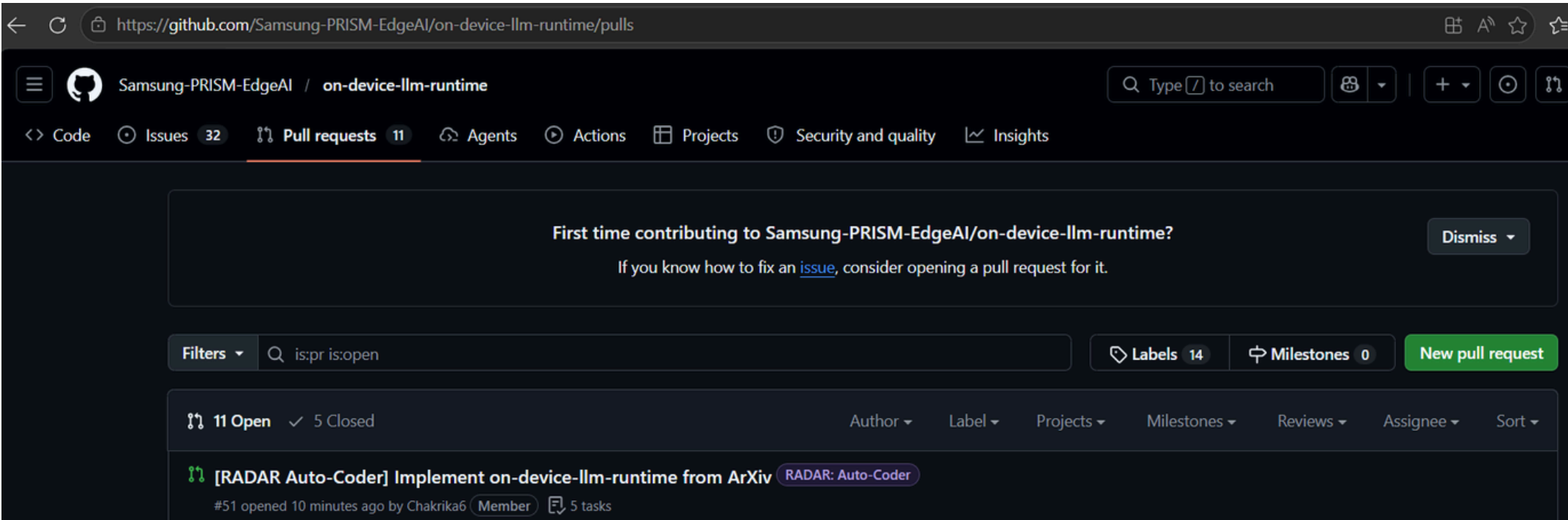
Zero-Downtime: Logs confirm a seamless auto-fallback from Gemini to Groq during API outages.

Uninterrupted Intelligence: Ensures the R&D pipeline never stalls, regardless of service availability.

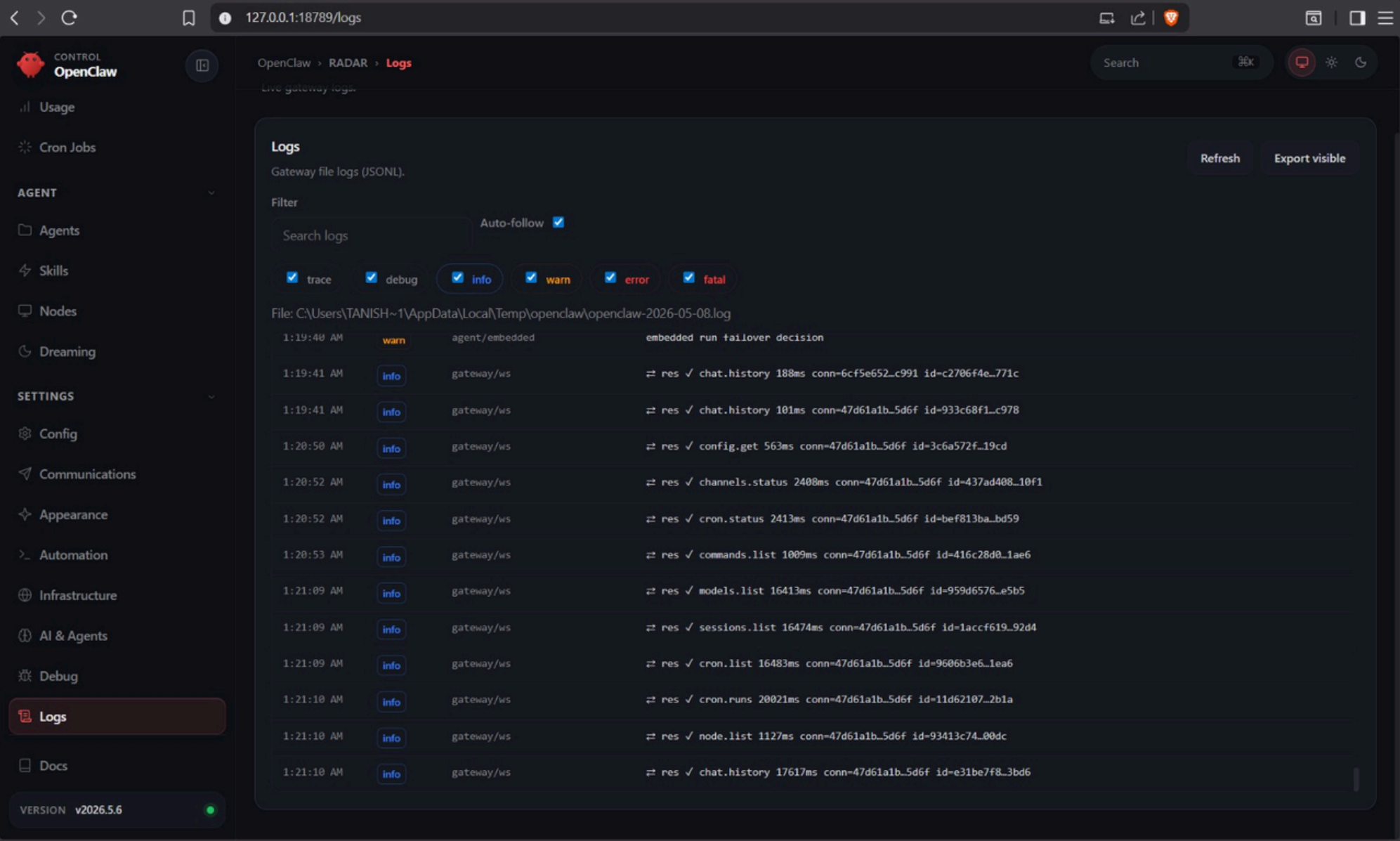
3. Strategic Briefing

High-Priority Scouting: Auto-generates GitHub Issues for competitive threats and macro-research.

Actionable Roadmaps: Research is filtered and prioritized based on its direct impact on Samsung’s goals.



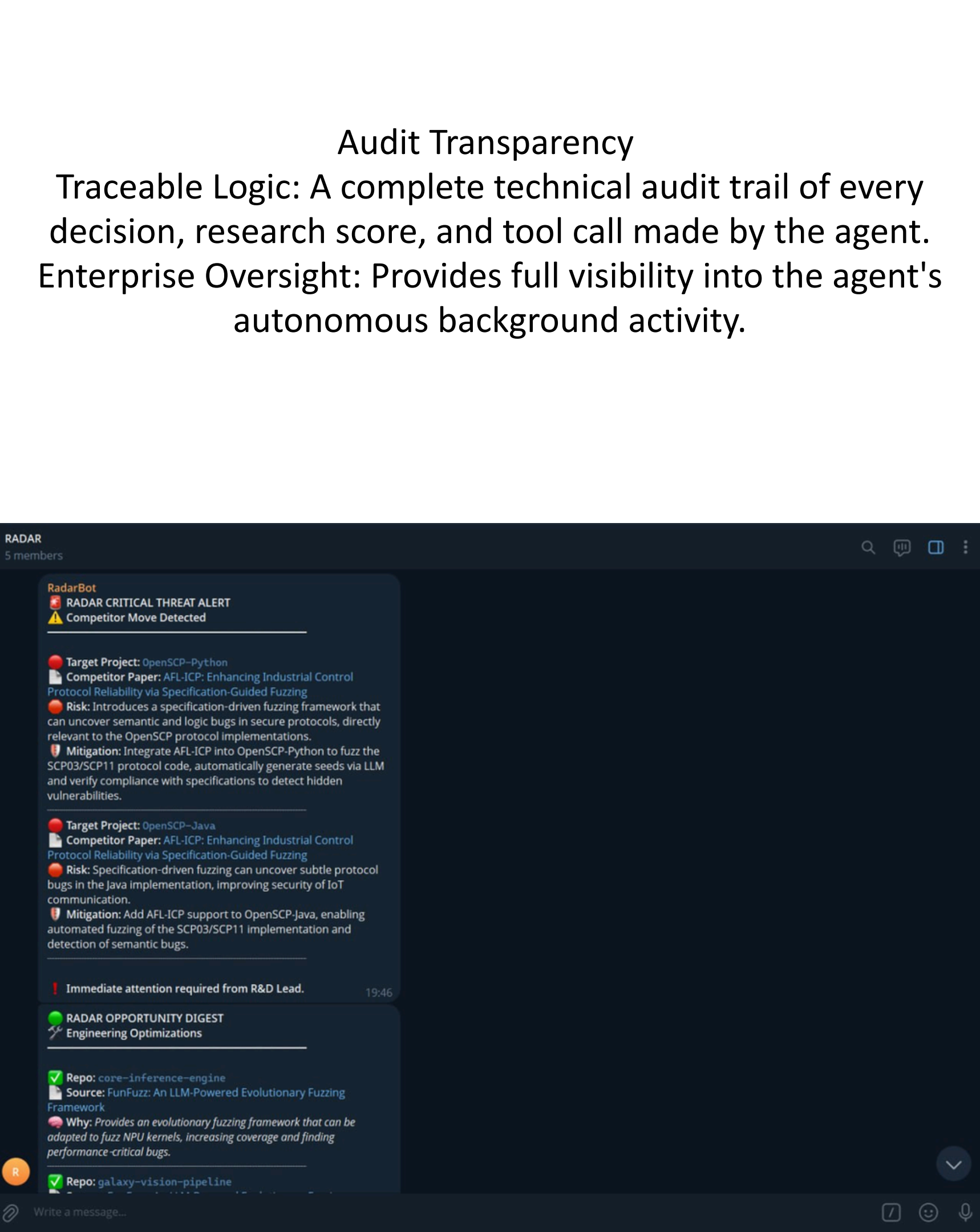


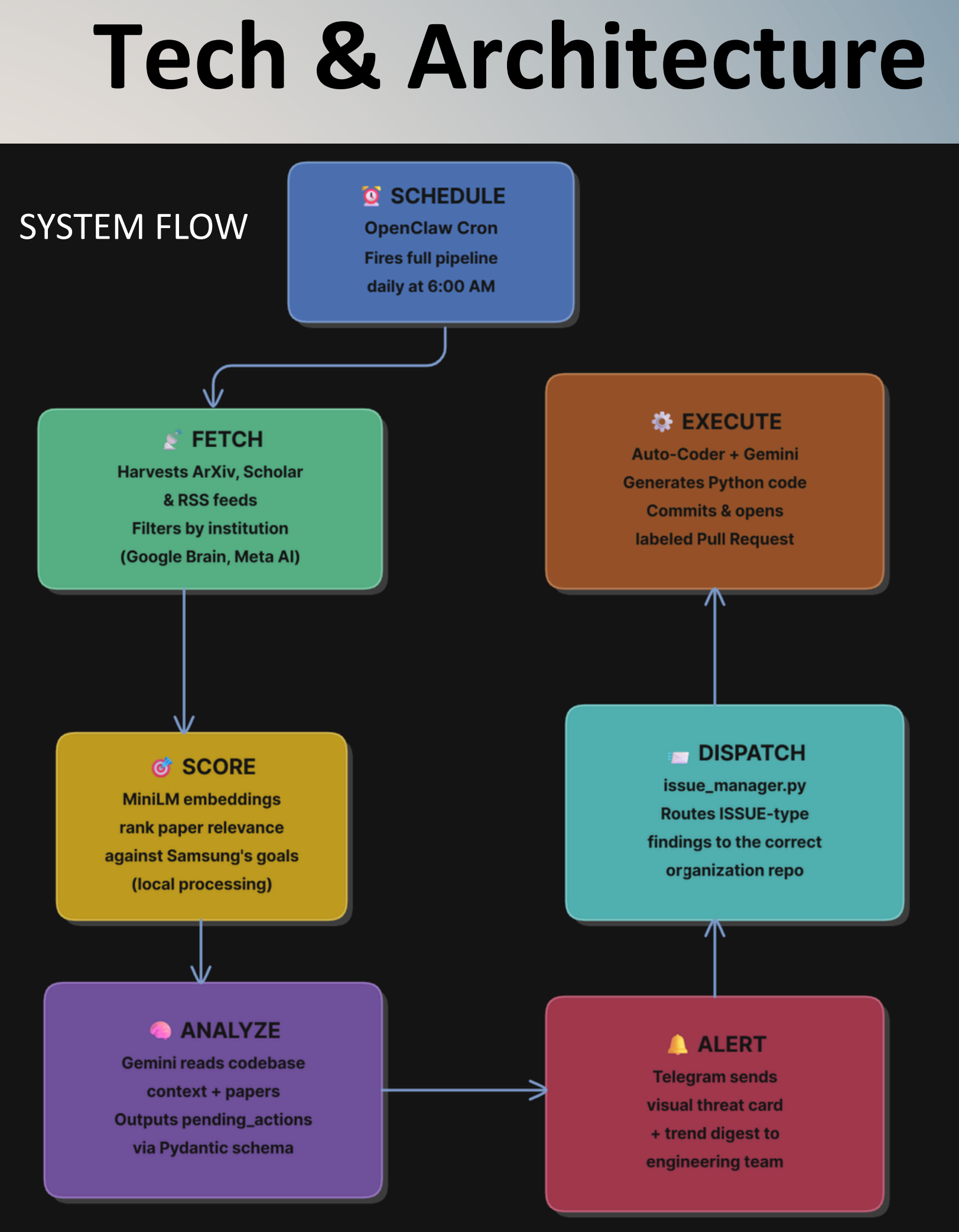
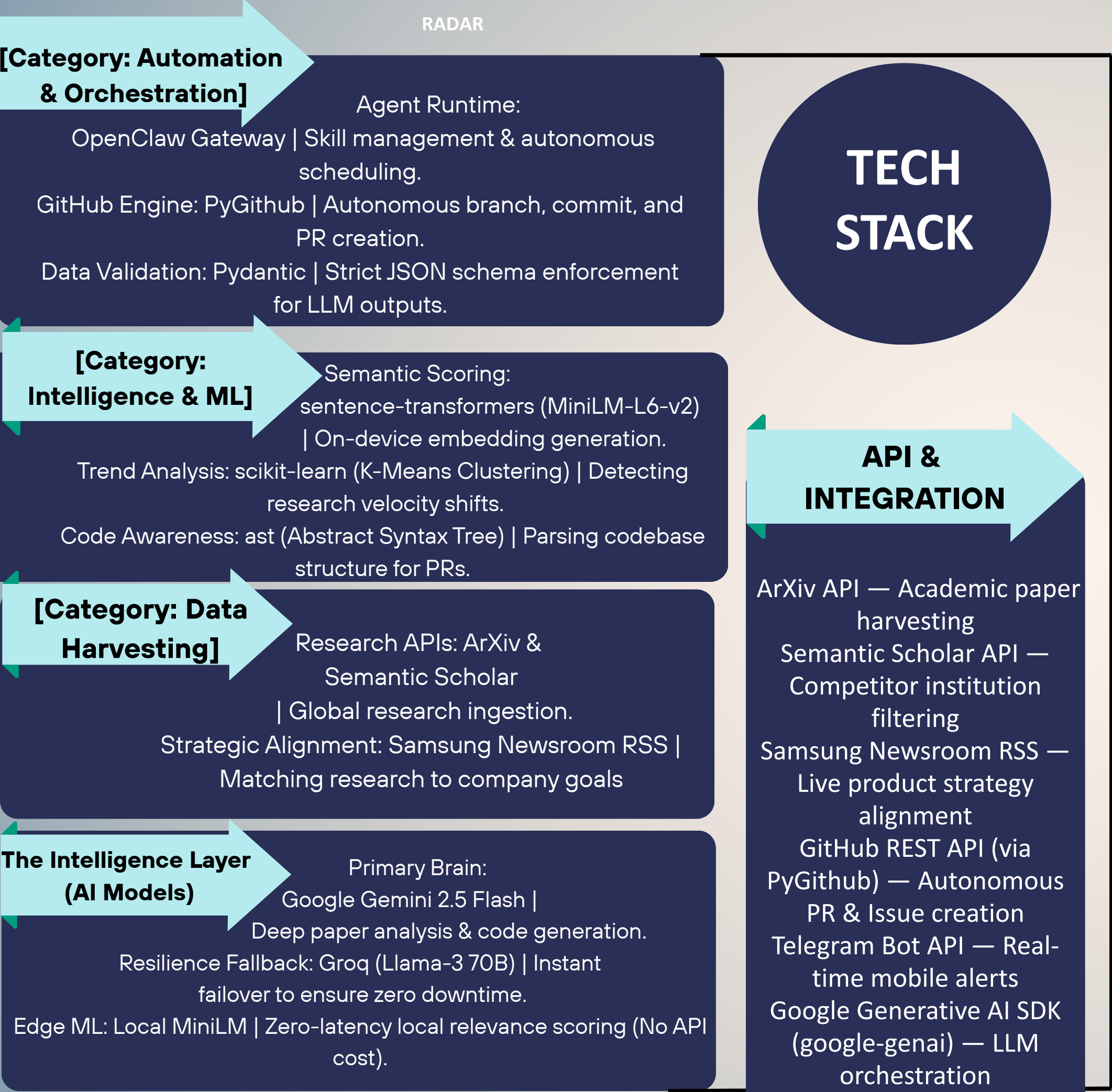


## Strategic Alerting

Critical Threat Alerts: Instant Telegram notifications for competitor breakthroughs (e.g., Apple, Meta) that impact Samsung’s projects.

Opportunity Digests: Research findings are mapped directly to specific repositories (e.g., core-inference) with clear, actionable mitigation steps.







# SCOPE

## Within Samsung:

### Multi-Division Deployment:

Seamlessly active across Galaxy AI, Bixby, SmartThings, and Semiconductor.

### Tailored Personas:

Custom SOUL.md configurations tuned to each division's specific engineering goals.

## Beyond RADAR:

### Org-Agnostic:

Adapts to any tech stack with GitHub repositories and research dependencies.

SaaS-Ready: Plug-and-play research automation for any enterprise.

### Broad Horizons:

Scalable to monitor patents, regulatory filings, and competitor product launches.

## By the Numbers:

### 40x Coverage:

Scanning 200+ papers daily vs. 5 manual reads  
Compressing multi-week research cycles to under 6 minutes.

Efficiency: One agent orchestrates the entire organization with zero headcount growth.

# Impact & Use cases

## Scenario 1

### The 8-Hour Breakthrough

Google Brain publishes at 11 PM; RADAR opens a working PR by 9 AM. Zero-touch research-to-code.

## Scenario 2

### Predictive Trend Spotting

K-Means clustering detects a surge in "On-Device Diffusion" papers. Real-time Telegram alerts drive preemptive sprint pivots.

## Scenario 3

### Cross-Repo Orchestration

One breakthrough triggers a PR in the NPU repo and an Issue in Camera AI. Single paper, multi-repo impact.

## Scenario 4

### Enterprise Resilience

Automated fallback from Gemini to Groq during API outages. Continuous execution with multi-model redundancy

# Differentiation

## MOAT

### AST-Driven Local Context .(The Integration Moat)

#### The Advantage:

RADAR uses Abstract Syntax Tree (AST) parsing to build a living graph of our private Samsung repositories

#### Why It's Defensible:

Targets exact functions (e.g., matching a new paper directly to `def load_npu_weights()`). it deeply maps our architecture, off-the-shelf competitors cannot replicate its relevance.

### Patent Proximity Scoring .(The Strategic Moat)

The Advantage: When competitor research drops (e.g., Google Brain), RADAR intercepts it instantly.

#### Why It's Defensible:

It autonomously cross-references live databases (Lens.org) to check for competitor patents. protecting Samsung from invisible macro threats.

## WHY RADAR STANDS OUT

### The Status Quo:

Existing tools (like ChatGPT or Semantic Scholar) stop at reading and summarizing PDFs. The engineer still has to figure out how to apply it.

### The RADAR Difference:

We aren't building a search engine; we built an execution engine. RADAR cross-references global research against our private repositories and autonomously writes the implementation code.

# Demo Link

GitHub Repo      [https://github.com/Sahitya3105/RIT\\_CodeRock](https://github.com/Sahitya3105/RIT_CodeRock)

Demo Link      IN GITHUB REPO ITSELF

Document submitted      IN GITHUB REPO ITSELF



# THE DIGITAL CTO

## Implementation "Vibe" Analysis

**The Feature**: Deep Source-Code Peeking.

**The Insight**: RADAR doesn't just read READMEs. It analyzes your inference.py and main.py files to understand the "Code Vibe"—the specific coding patterns and library versions your team actually uses. It ensures the PRs it writes are not just "correct," but "consistent" with your team's style.

## Problem-Aware Prioritization (Issue Scanning)

**The Feature**: Active Pain-Point Discovery.

**The Insight**: RADAR scans your Open GitHub Issues. If a developer logs a performance bug (e.g., "Inference latency too high in NPU layer"), RADAR automatically prioritizes research papers that address that specific issue in the next 6:00 AM run. It doesn't just find research; it finds solutions to your current problems.

## Zero-Touch "SaaS-Ready" Onboarding

**The Feature**: Autonomous Strategy Synthesis.

**The Insight**: There is no manual configuration for new projects. Point RADAR at a GitHub Organization, and its org\_scanner self-discovers the entire company's research roadmap. It builds its own "Strategy Document" (org\_profile.json) like a human CTO would.

## Privacy-First "IP Shield"

**The Feature**: Local Architectural Mapping.

**The Insight**: To protect Samsung's Intellectual Property, the deep code analysis (AST Parsing) happens entirely locally. We never upload your raw codebase to the cloud. We only send the "Architectural Graph" to the LLM, ensuring that your core IP remains secure within your private environment.